

NeuroAnalyzer report: ROI / Image Analysis

NeuroAnalyzer v0.3.0 · MATLAB 2026a · 2026-09-25T03:21:36+00:00 · window image: exportapp

ROI / Image Analysis

Motion correction, multiple ROIs and cell detection, $\Delta F/F$, flow speed, kymograph and vessel diameter

Help

96 X 96 px · 150 frames · grayscale
loaded to 14.8 · roiMask (113 px)

2 Preprocess (optional)

- ☐ Motion correction (input)
- ☐ BW 256 levels
- ☐ Smooth (Gaussian, sigma 2 px)
- ☐ Normalize each frame to 0-1

Applied in this order when you click Run.

3 ROIs and line

Add ROI

Detect cells

Draw line

Clear ROIs and line

ROI: ROI 1 (113 px), Line: 30 px long.

4 Analysis

Method ΔF/F (gCaMP)

ΔF/F baseline (frames) 30

☐ Robust diameter (ignore blood cells)

Run

5 Export

Export results

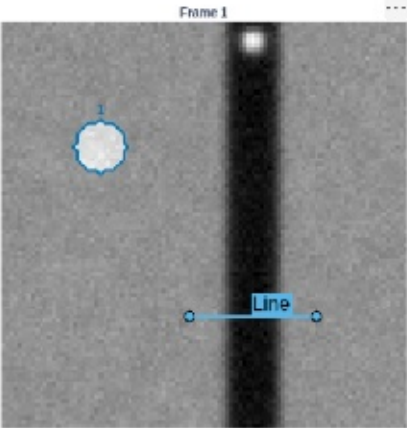
Ready to export ΔF/F (gCaMP)

Save session...

Open session...

Report (PDF)...

Image, ROIs and line



ROIs (double click a name to rename)

V	Name	Area px	Source
1	ROI 1	113	mask

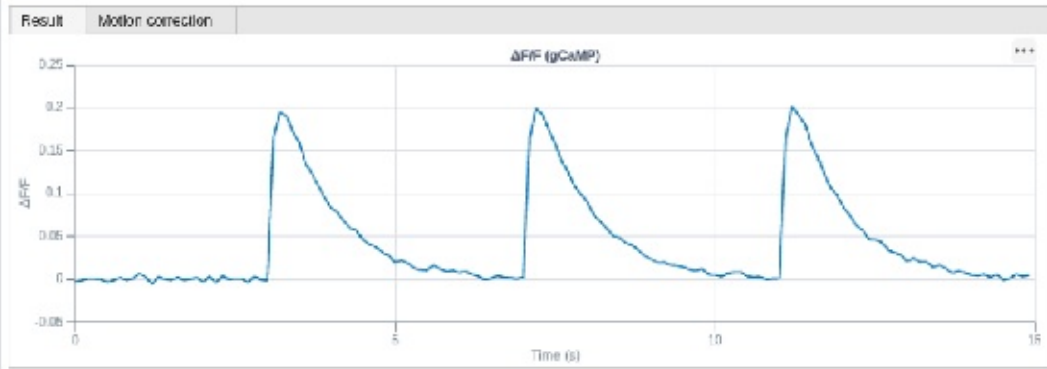
Remove ROI

Show First frame

Report with correlation

0

Results



Session saved to ROIAnalysisApp.nasession.mat (1 Input file(s) with MDO, settings and results).

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Window: ROI / Image Analysis (ROIAnalysisApp)

NeuroAnalyzer v0.3.0
MATLAB 26.1.0.3346908 (R2026a) Update 5
OS: Linux 6.17.0-1022-azure (GLNXA64)
Session created: 2026-09-25T03:21:36+00:00
Report created: 2026-09-25T03:21:36+00:00

NOTES

ROIAnalysisApp walkthrough: demo data

INPUTS (1)

1. Image stack: demo_imaging.mat
- path: /tmp/NeuroAnalyzerDemo/demo_imaging.mat
- 4,685,638 bytes, modified 2026-09-25T03:17:39
- MD5: 25b58ba39029b52115e00c93ca22f582

SETTINGS

generator = ''
preprocess.motionCorrection = false
preprocess.bw256 = false
preprocess.smooth = false
preprocess.normalize = false
method = ΔF/F (gCaMP)
dfffBaselineFrames = 30
robustDiameter = false
display = First frame
detectThresholdField = 0
detectThresholdUsed = []
roiNames = {ROI 1}
roiSources = {mask}
roiPositions = [1x1 cell]
roiMasks = [96x96 logical]
line = [45 70;75 70]

KEY RESULTS

Stack: 96 x 96 px, 150 frames; 1 ROI(s); line [45 70;75 70] px
Method: ΔF/F (gCaMP) over 150 frames
dF/F ROI 1: mean 0.04123, max 0.2012 at t = 11.2